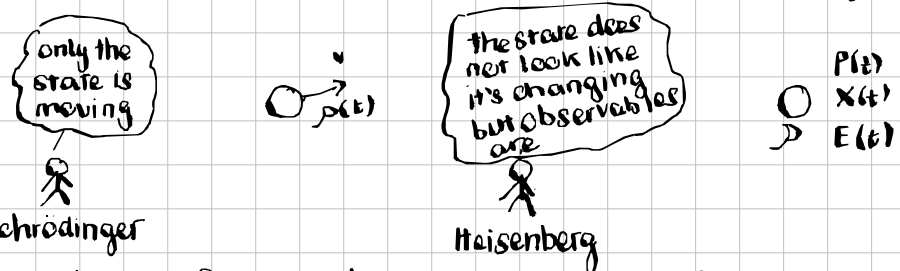
Quantum Pictures

Activator: Quantum pictures arise from our freedom to choose different reference frames in Hilbert space. The physical evolution remains the same, but what appears to evolve depends on the frame from which we describe the system.

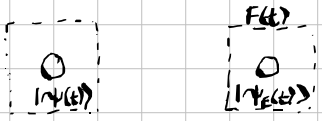
One of the early confusions for a quantum scientists is the many different pictures.
The concept is the same as the reference frame in relativity.



There is also a frame where a person sees the state and observables are changing with time.
This case is called the interaction picture.

The what: "state" $|\psi\rangle$ or $p(t)$ and "physical observable" $H, p, x \ldots$

The freedom: $\hat{F}(t)$ is the reference frame that we describe the system



The freedom gives a relative energy

$$\hat{H}_E = \hat{F} H \hat{F}^\dagger - i\hbar \hat{F} \frac{\partial}{\partial t} \hat{F}^\dagger$$

The math shows

$$i\frac{d}{dt}|\psi(t)\rangle = H(t)|\psi(t)\rangle \quad \text{and} \quad i\frac{d}{dt}|\psi_E\rangle = H_E(t)|\psi_E\rangle$$

That is $$i\frac{d}{dt}|\psi_E\rangle = \left(\hat{F} H \hat{F}^\dagger - i\hbar \hat{F}\frac{\partial}{\partial t}\hat{F}^\dagger\right)|\psi_E\rangle \qquad \text{Let } |\psi\rangle = \hat{F}^\dagger|\psi_E\rangle$$

$$\frac{d}{dt}|\psi_E\rangle = \frac{d\hat{F}}{dt}|\psi\rangle + \hat{F}\frac{d}{dt}|\psi\rangle$$

The freedom allows us to go into any reference

Activator: In quantum optics, two common choices of rotating frame allow us to simplify quantum dynamics: one removes the free evolution to isolate the interaction, while the other removes the common carrier frequency oscillation to reveal the relative frequencies

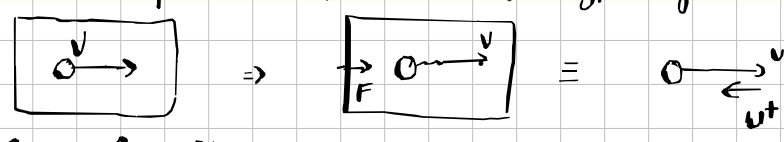
Example $\hat{H} = \frac{\hbar\omega_0}{2}\sigma_z + \hbar\omega a^\dagger a + \hbar g(\sigma^\dagger + \sigma^-)(a^\dagger + a)$

The free energy term can be removed relatively since its energy is fixed.

By doing this,

$$H_0 = \frac{\hbar\omega_0}{2}\sigma_z + \hbar\omega a^\dagger a \quad \text{and} \quad F = e^{iH_0 t/\hbar} = U^\dagger$$

The reason for $F = U^\dagger$ is simple. F is the frame and U typically describe the evolution



Heisenberg is when $F = V = U^\dagger$

Therefore, $\hat{H}_E = \hat{F}\hat{H}\hat{F}^\dagger - i\hbar\hat{F}\partial_t\hat{F}^\dagger$

$\hat{F}\sigma^+\hat{F}^\dagger = e^{i\omega_0 t}\sigma^+$ $\quad \hat{F}\sigma^-\hat{F}^\dagger = e^{-i\omega_0 t}\sigma^-$ $\quad F H_0 F^\dagger = H_0$

$\hat{F} a \hat{F}^\dagger = e^{-i\omega t} a$ $\quad \hat{F} a^\dagger \hat{F}^\dagger = e^{i\omega t} a^\dagger$

Note

$$\hat{F} a \hat{F}^\dagger = a(t) \Rightarrow \text{Heisenberg } \frac{d}{dt}\hat{a}(t) = i[H_0, \hat{a}(t)] = -i\omega_0 \hat{a}$$

The second term

$-i\hbar\hat{F}\partial_t\hat{F}^\dagger = -H_0$ cancels the H_0

So effectively,

$$\hat{H}_E = \hbar g\left[\sigma^+ e^{i\omega_0 t} + \sigma^- e^{-i\omega t}\right]\left(a^\dagger e^{i\omega t} + a e^{-i\omega t}\right)$$

Under RWA

$$\hat{H}_E = \hbar g\left[\sigma^+ a e^{i\Delta t} + \sigma^- a^\dagger e^{i\Delta t}\right]$$

If originally, we rotate as

$$H_0' = \frac{\hbar\omega}{2}\sigma_z + \hbar\omega a^\dagger a$$

The second term does not fully cancel σ_z term which

$$\hat{H}_E' = \frac{\hbar\Delta}{2}\sigma_z + \hbar g(\sigma^+ a + \sigma^- a^\dagger)$$

Both are often used in quantum mechanics

Example $H_{tot} = H_{sys} + H_{Bath} + H_{int}$

The bath is written as

$$H_{Bath} = \int_0^\infty d\omega\, \omega\, b^\dagger(\omega) b(\omega)$$

Interaction becomes

$$H_{int} = i\int_0^\infty d\omega\, k(\omega)[b(\omega) + b^\dagger(\omega)][c - c^\dagger] \qquad i \text{ is caused by definition of } c$$

Going to frame $H_0 = H_{sys} + H_{bath}$

$$H_E = i\int_0^\infty d\omega\, k(\omega)\left[b^\dagger c\, e^{i(\omega-\Omega)t} - c^\dagger b\, e^{-i(\omega-\Omega)t}\right]$$

The interaction is strong near Ω, thus it is safe to extend $-\infty \to \infty$

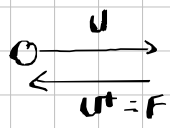
Similiar as above when the bath also rotates as Ω, then

$$H_E' = \int_0^\infty d\omega\,(\omega - \Omega)\, b^\dagger(\omega) b(\omega) + i\sqrt{\frac{\gamma}{2\pi}}\int_{-\infty}^\infty d\omega\left[c b^\dagger - c b\right]$$

Heisenberg Picture

The Heisenberg is exactly when

$$\hat{F}(t) = U^{\dagger}(t)$$



We can arrive at the equation from previous solutions

That is $|\psi\rangle = U|\psi_E\rangle$ (U removes the frame)

$|\psi_E\rangle = U^{\dagger}U|\psi(0)\rangle$ $\Rightarrow$ stationary

The whole dynamics is carried over to moving frame.

$$\hat{H}_E = 0 = FHF^{\dagger} - i\hbar \dot{F}F^{\dagger} \Rightarrow \dot{F}F^{\dagger} = \frac{1}{i\hbar} FHF^{\dagger}$$

Observe that

$$\hat{A}_H = FA_SF^{\dagger} \Rightarrow \frac{d\hat{A}_H}{dt} = \dot{F}A_SF^{\dagger} + F\frac{\partial A_S}{\partial t}F^{\dagger} + FA_S\dot{F}^{\dagger}$$

Since $FF^{\dagger} = I \Rightarrow \dot{F}F^{\dagger} = -F\dot{F}^{\dagger}$

Therefore

$$\dot{F}F^{\dagger} = \frac{1}{i\hbar}\hat{H}_H \Rightarrow \frac{d\hat{A}_H}{dt} = -\frac{1}{i\hbar}H_H\hat{A}_H + \frac{1}{i\hbar}\hat{A}_H\hat{H}_H + F\frac{\partial \hat{A}}{\partial t}F^{\dagger}$$

We obtain:

$$\frac{dA_H}{dt} = \frac{1}{i\hbar}[\hat{H}_H, \hat{A}_H] + \left(\frac{\partial \hat{A}_S}{\partial t}\right)_H \qquad \text{with } H_H = H \text{ since } [H, e^{-iHt}] = 0$$

The fact that $\hat{A}_H = F\hat{A}_SF^{-1}$ can be seen in a different context

$$\langle A(t)\rangle = Tr\{A(t)\rho(t)\} = Tr\{AU\rho(t_0)U^{\dagger}\} = Tr\{U^{\dagger}AU\rho(t_0)\}$$

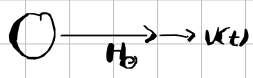
Interaction Picture

Instead of choosing a reference frame that follows the entire Hamiltonian, we choose one that follows only part of it. This is often nice when the energy level has the form

$$\hat{H} = \hat{H}_0 + \hat{V}(t)$$

We move the frame to remove the fixed energy

$$F(t) = e^{i\hat{H}_0 t/\hbar}$$



This gives

$$\hat{H}_E = F\left(\hat{H}_0 + \hat{V}(t)\right)F^\dagger - i\hbar \dot{F}F^\dagger = F\hat{H}_0F^\dagger + F\hat{V}F^\dagger - F\hat{H}_0F^\dagger = F\hat{V}F^\dagger$$

The interaction state is simply

$$|\psi_E\rangle = F|\psi\rangle \Rightarrow i\hbar\frac{d}{dt}|\psi_E(t)\rangle = V_E|\psi_E(t)\rangle$$

In compare with

$$i\hbar\frac{d}{dt}|\psi\rangle = (H_0 + V)|\psi\rangle$$

Similiarly, the equation of motion is

$$\hat{A}_I = FA_SF^\dagger$$

$$\frac{d\hat{A}_E}{dt} = \dot{F}\hat{A}_SF^\dagger + F\partial_t\hat{A}_SF^\dagger + FA_SF^\dagger \qquad \text{using } \dot{F} = \tfrac{1}{i\hbar}\hat{H}_0F \text{ and } \dot{F}^\dagger = -\frac{i}{\hbar}F^\dagger H_0$$

$$= \tfrac{i}{\hbar}[\hat{H}_0, \hat{A}_E] + \left(\frac{\partial A_S}{\partial t}\right)_I$$

The state evolves

$$i\hbar\frac{d}{dt}|\psi_E\rangle = V_E|\psi_E\rangle$$